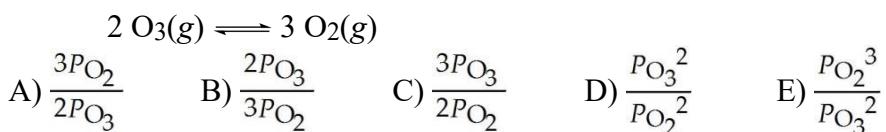


Chapter 15 Homework

Part A Multiple choice (Select the one that is best in each case. 1 point/question)

1. Which of the following statements is true in a reaction system at equilibrium?
- A) The equilibrium constant is zero.
 - B) The number of collisions per unit time between reactants is equal to the number of collisions per unit time between products.
 - C) Reactants are reacting to form products at the same rate as products are reacting to form reactants.
 - D) Reactants and products are present in equimolar amounts.
 - E) The product of the concentrations of the products divided by the product of the concentrations of the reactants is always a constant.

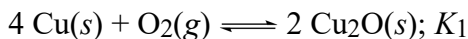
2. The equilibrium expression for K_p for the reaction below is _____.



3. For the reaction $2 \text{H}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{H}_2\text{O}(\text{g})$, what is the relationship between K_c and K_p at temperature T ?

- A) $K_p = K_c(RT)^2$
- B) $K_c = K_p(RT)$
- C) $K_c = K_p$
- D) $K_p = K_c(RT)$
- E) $K_c = K_p(RT)^2$

4. Given the equilibrium constants for the following reactions:



What is K for the system $2 \text{Cu}(\text{s}) + \text{O}_2(\text{g}) \rightleftharpoons 2 \text{CuO}(\text{s})$?

- A) $(K_2)^2/(K_1)$
- B) $K_1 \times K_2$
- C) $(K_1)(K_2)^{1/2}$
- D) $(K_1)^{1/2}/(K_2)$
- E) $(K_2)^{1/2}/(K_1)$

5. Exactly 1.0 mol N_2O_4 is placed in an empty 1.0-L container and allowed to reach equilibrium described by the equation $\text{N}_2\text{O}_4(\text{g}) \rightleftharpoons 2 \text{NO}_2(\text{g})$.

If at equilibrium the N_2O_4 is 28.0% dissociated, what is the value of the equilibrium constant, K_c , for the reaction under these conditions?

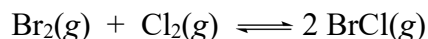
- A) 0.44 B) 2.3 C) 0.31 D) 0.78 E) 0.11

6. Which of the following statements is true?

- A) When two opposing processes are proceeding at identical rates, the system is at equilibrium.
- B) An endothermic reaction shifts toward reactants when heat is added to the reaction.
- C) The concentration of the products equals that of the reactants and is constant at equilibrium.
- D) Catalysts are an effective means of changing the position of an equilibrium.
- E) None of the above statements is true.

Chapter 15 Homework

7. At 400 K, the equilibrium constant for the reaction



is $K_p = 7.0$. A closed vessel at 400 K is charged with 1.00 atm of $\text{Br}_2(g)$, 1.00 atm of $\text{Cl}_2(g)$, and 2.00 atm of $\text{BrCl}(g)$. Use Q to determine which of the statements below is true.

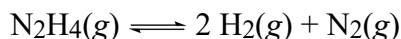
- A) The equilibrium partial pressures of Br_2 , Cl_2 , and BrCl will be the same as the initial values.
- B) The equilibrium partial pressure of Br_2 will be greater than 1.00 atm.
- C) At equilibrium, the total pressure in the vessel will be less than the initial total pressure.
- D) The equilibrium partial pressure of BrCl will be greater than 2.00 atm.
- E) The reaction will go to completion since there are equal amounts of Br_2 and Cl_2

8. Consider the reaction represented by the equation

$\text{N}_2(g) + 3\text{H}_2(g) \rightleftharpoons 2\text{NH}_3(g)$. What happens to the equilibrium position when an inert gas is added to this system (as represented above) at equilibrium?

- A) If the container is rigid, the equilibrium position shifts. If the container is fitted with a movable piston, nothing happens to the equilibrium position.
- B) If the container is rigid, nothing happens to the equilibrium position. If the container is fitted with a movable piston, the equilibrium position shifts.
- C) Nothing happens to the equilibrium position no matter what the container is like.
- D) The equilibrium position shifts no matter what the container is like.
- E) The value of the equilibrium constant must be known to answer this question.

9. Consider the decomposition of hydrazine as shown below.



At a certain temperature, $K_p = 2.5 \times 10^3$. When pure hydrazine is placed in an empty vessel at this temperature, equilibrium is reached after 30.0% of the hydrazine has decomposed. Calculate the partial pressure of hydrogen gas at equilibrium.

- A) 76 atm B) 5776 atm C) 54 atm D) 127 atm E) none of these

10. The value of the equilibrium constant K is dependent on:

- I. the temperature of the system.
 - II. the nature of the reactants and products.
 - III. the concentration of the reactants.
 - IV. the concentration of the products.
- A) I and II only
B) III and IV only
C) II and III only
D) three of these
E) none of these

11. Consider the following equilibrium:



Which of the following statements about the equilibrium is false?

- A) Removing HI as it forms forces the equilibrium to the right.
- B) If the system is cooled, the left side is favored.
- C) This is a heterogeneous equilibrium.
- D) If the pressure on the system is decreased by changing the volume, the right side is favored.
- E) Adding more $\text{I}_2(g)$ increases the equilibrium constant.

Chapter 15 Homework

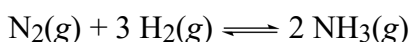
12. Consider the following equilibrium:



Which of the following statements is true?

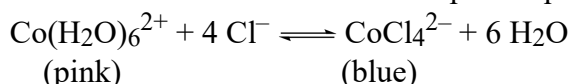
- A) If the reaction mixture initially contains only $\text{OF}_2(\text{g})$, then at equilibrium, the reaction mixture will consist of essentially only $\text{O}_2(\text{g})$ and $\text{F}_2(\text{g})$.
- B) For this equilibrium, $K_c = K_p$.
- C) If the reaction mixture initially contains only $\text{OF}_2(\text{g})$, then the total pressure at equilibrium will be less than the total initial pressure.
- D) If the reaction mixture initially contains only $\text{O}_2(\text{g})$ and $\text{F}_2(\text{g})$, then at equilibrium, the reaction mixture will consist of essentially only $\text{OF}_2(\text{g})$.
- E) If the reaction mixture initially contains only $\text{O}_2(\text{g})$ and $\text{F}_2(\text{g})$, then the total pressure at equilibrium will be greater than the total initial pressure.

13. What effect will spraying liquid water into the equilibrium given below have if NH_3 is far more soluble in water than is N_2 or H_2 ?



- A) More $\text{NH}_3(\text{g})$ will form.
- B) More $\text{H}_2(\text{g})$ will form.
- C) Less $\text{NH}_3(\text{g})$ will form.
- D) This will not affect the system.
- E) More $\text{N}_2(\text{g})$ will form.

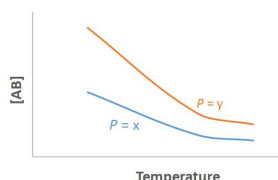
14. When cobalt chloride is added to pure water, the Co^{2+} ions hydrate. The hydrated form then reacts with the Cl^- ions to set up the equilibrium shown here:



Which statement accurately describes the change that the system will undergo if water is added?

- A) The color will become more blue.
- B) The equilibrium will shift to the right.
- C) More water will be produced.
- D) More chloride ions will be produced.
- E) There will be less of the hydrated cobalt ion at the new equilibrium position.

15. The following graph represents the yield of the compound AB at equilibrium in the reaction $\text{A}(\text{g}) + \text{B}(\text{g}) \rightleftharpoons \text{AB}(\text{g})$ at two different pressures, x and y , as a function of temperature.



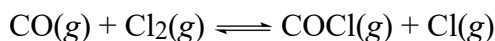
According to the above information, which of the following statements is true?

- A) The reaction is endothermic and $x > y$.
- B) The reaction is endothermic and $x < y$.
- C) The reaction is exothermic and $x < y$.
- D) The reaction is exothermic and $x > y$.
- E) The reaction is exothermic and $x = y$.

Chapter 15 Homework

Part B Short questions (15 points)

16. (3 points) Both the forward reaction and the reverse reaction in the following equilibrium are believed to be elementary steps:



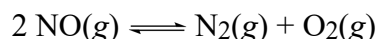
At 25 °C, the rate constants for the forward and reverse reactions are

$1.4 \times 10^{-28} \text{ M}^{-1} \text{ s}^{-1}$ and $9.3 \times 10^{10} \text{ M}^{-1} \text{ s}^{-1}$, respectively.

(a) What is the value for the equilibrium constant at 25 °C

(b) Are reactants or products more plentiful at equilibrium? Explain.

17. (4 points) At 2000 °C, the equilibrium constant for the reaction



is $K_c = 2.4 \times 10^3$. If the initial concentration of NO is 0.175 M, what are the equilibrium concentrations of NO, N₂, and O₂?

18. (4 points) At 900 °C, $K_c = 0.0108$ for the reaction $\text{CaCO}_3\text{(s)} \rightleftharpoons \text{CaO(s)} + \text{CO}_2\text{(g)}$. A mixture of CaCO₃, CaO, and CO₂ is placed in a 10.0-L vessel at 900 °C. For the following mixtures, will the amount of CaCO₃ increase, decrease, or remain the same as the system approaches equilibrium?

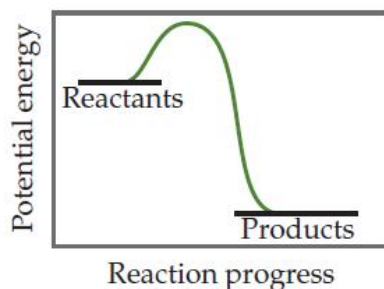
(a) 30.0 g CaCO₃, 15.0 g CaO, and 4.25 g CO₂

(b) 1.50 g CaCO₃, 25.0 g CaO, and 5.66 g CO₂

19. (4 points)

(a) Based on the following energy profile, predict whether $k_f > k_r$ or $k_f < k_r$. Explain.

(b) Suppose that both the forward reaction and the reverse reaction are elementary reactions, predict whether the equilibrium constant for the process is greater than 1 or less than 1. Explain.



Chapter 15 Homework Answer Sheet

Name:

Student ID:

Instructor:

Score:

Part A Multiple choice (15 points)

1-5_____

6-10_____

11-15_____

Part B Short questions (15 points)